TELECOM SERVICE ASSURANCE &
COST-TO-SERVE ANALYTICS PLATFORM

Incident Intelligence for Service Assurance, Cost Optimization, and SLA Performance

BUSINESS REQUIREMENTS DOCUMENT

Technology Stack: SQL (MySQL) · Python · Power BI (Data Modeling, DAX, Power Query)

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1. Executive Summary

Broadband service assurance operations generate incident volume that varies constantly in nature and cost, but is rarely reviewed as one connected dataset. Incident records, SLA outcomes, dispatch decisions and their costs typically sit in separate operational views, which makes it difficult to see which fault types or resolution strategies actually drive cost and SLA risk, as opposed to which simply generate the most tickets.

This project analyzes 25,000 broadband service incidents recorded across 650 sites in 18 states between January 2023 and December 2025, using SQL for incident-level analysis and Power BI for a six-page reporting layer covering root cause, resolution efficiency, cost and business recommendations. The objective was to identify where cost-to-serve and SLA risk actually concentrate, and whether that concentration matches incident volume.

It does not. Fiber Team Dispatch, Vendor Support and NOC Support together account for 41.8% of incidents but 79.6% of the ₹310.36 Crore total cost-to-serve. Escalated incidents breach SLA at 53.27%, against 1.11% for incidents that are not escalated. These patterns, and the fault categories and site segments behind them, are the basis for the recommendations in this document.

2. Business Problem

NOC and field operations resolve broadband service incidents one at a time, without a systematic view of which fault types or resolution choices consume a disproportionate share of cost, SLA risk or repeat-visit exposure across the network.

Total cost-to-serve for the incidents in this dataset is ₹310.36 Crore. Average operational cost per incident (dispatch, labor, vendor support) is ₹1,487.76, while average service impact cost per incident is ₹1,22,656.77 — over 80 times higher. Cost concentrates in specific resolution strategies and fault categories, but company-wide averages hide that concentration.

The overall SLA breach rate is 9.99%, against an internal target of 6%. Optical Network and Access Equipment incidents alone account for 82.7% of all SLA breaches while representing 45.1% of incident volume. Escalated incidents breach SLA at 53.27%, nearly 48 times the 1.11% rate for incidents that are not escalated.

The challenge is not reducing incident volume alone, but prioritizing the incidents that drive the highest service disruption, SLA risk, and operational cost.

3. Data Overview

The analysis is built on a star-schema data model: one fact table of incident records joined to four dimension tables covering date, site, fault and vendor/technology.

- Analysis period: January 1, 2023 – December 31, 2025 (3 full years)
- Incident volume: 25,000 broadband service assurance incidents
- Geographic coverage: 650 sites across 18 states and union territories, grouped into 4 zones (Central, East, North, West)
- Site segmentation: 3 site types (Urban, Semi-Urban, Rural) and 3 criticality tiers (Tier 1–3)
- Technology scope: 4 vendors (Huawei, Nokia, ZTE, Tejas) across 3 access technologies (GPON, XGS-PON, BRAS)
- Fault taxonomy: 9 fault categories covering 40 fault sub-types

- Technology Stack: Python (Data Generation & ETL) · SQL (MySQL) · Power BI (Data Modeling, DAX, Power Query & Reporting)

4. Stakeholders

- NOC Operations Managers — resolution strategy selection drives 79.6% of total cost-to-serve.
- Field Operations & Dispatch Planning — Fiber Team Dispatch and Field Visit together account for 96.9% of all SLA breaches.
- SLA & Customer Experience Owners — escalation status, not fault severity, is the strongest observed predictor of SLA breach (53.27% vs. 1.11%).
- Vendor Management — Vendor Support costs an average of ₹4,17,953 per incident, the highest of any resolution path.
- Network Planning & Infrastructure Investment — Optical Network and Access Equipment jointly drive the majority of both cost and SLA risk.

5. Business Questions

Q1 — Cost Concentration: Fiber Team Dispatch, Vendor Support and NOC Support together handle 41.8% of incidents (10,438 of 25,000) but account for 79.6% of total cost-to-serve (₹246.93 Crore of ₹310.36 Crore). Is incident volume the right basis for deciding where to invest in prevention, or does cost concentrate somewhere else entirely?

Q2 — SLA Risk Concentration: Optical Network and Access Equipment incidents represent 45.1% of volume but 82.7% of all SLA breaches, and within Optical Network alone, close to 1 in 5 incidents (19.74%) breaches SLA. Are these two categories resourced in line with the risk they carry?

Q3 — Escalation and SLA Outcome: Escalated incidents breach SLA at 53.27%, against 1.11% for incidents that are not escalated. Is escalation being used as an early-warning signal for proactive intervention, or only as a status label applied after an incident is already at risk?

Q4 — Resolution Strategy Cost Efficiency: Remote Resolution costs an average of ₹515 per incident; Hardware Replacement costs ₹4,21,540 — over 800 times more. Vendor Support alone accounts for 21.3% of total cost-to-serve while used in only 6.3% of incidents. Is resolution strategy chosen based on what a fault needs, or on availability and habit?

6. Analytical Requirements

AR1 (→Q1): Aggregate total incident cost (operational + service impact) and incident count by resolution strategy; rank each by cost share and separately by volume share to identify strategies where cost concentration exceeds volume share.

AR2 (→Q2): Aggregate SLA breach count and breach rate by fault category (9 categories); compare each category's share of total incident volume against its share of total breaches.

AR3 (→Q3): Segment incidents by escalation flag; compare SLA breach rate and incident count between escalated and non-escalated incidents.

AR4 (→Q4): Aggregate average operational cost, service impact cost and total cost per incident by resolution strategy; compare each strategy's share of incidents against its share of total cost-to-serve.

7. Dashboard Requirements

Dashboard Page	Analytical Purpose
Executive Overview	Incident, SLA, dispatch and cost KPIs; Pareto view of which fault categories drive incident volume; incident volume vs. SLA performance trend against the internal SLA target.
Operational Root Cause Analysis	Vendor performance comparison, geographic distribution of incidents, and operational cost concentration by fault category.
Resolution & Efficiency Analysis	Resolution time trend over the analysis period, resolution driver breakdown, and resolution time distribution.
Financial & Cost Analytics	Cost trend over time, cost concentration by fault category and vendor, and Pareto view of cost drivers.
Business Insights & Strategic Recommendations	Narrative summary of key findings and recommended actions, tied directly to the preceding analysis pages.
Incident & Cost Investigation	Drillthrough detail view for any fault category, showing incident-level cost, geography and resolution detail behind a headline number.

8. KPI Definitions

KPI	Business Definition	Dashboard Page
Total Incidents	Count of all logged broadband service assurance incidents in the analysis period.	Executive Overview
SLA Breach Rate	Share of incidents that exceeded their severity-based resolution time target.	Executive Overview
SLA Target	Internal benchmark ceiling for SLA breach rate.	Executive Overview
Dispatch Rate	Share of incidents that required a physical field visit or vendor dispatch to resolve.	Investigation
Total Cost-to-Serve	Combined operational cost (dispatch, labor, vendor support) and service impact cost (customer / revenue impact) across all incidents.	Financial & Cost Analytics
Average Cost per Incident	Total cost-to-serve divided by total incident count.	Financial & Cost Analytics
Repeat Fault Rate	Share of incidents that recurred at the same site for the same fault within the defined detection window.	Resolution & Efficiency
Escalation Rate	Share of incidents escalated beyond first-line resolution.	Business Insights

KPI	Business Definition	Dashboard Page
Pareto %	Cumulative share of cost or incident volume contributed by fault categories, ranked highest to lowest.	Executive Overview

9. Key Discoveries

Discovery 1 — Cost concentrates in three resolution strategies, not in incident volume.

Resolution Strategy	Share of Incidents	Share of Total Cost
Fiber Team Dispatch	22.1%	40.08%
Vendor Support	6.3%	21.28%
NOC Support	13.3%	18.20%

Fiber Team Dispatch, Vendor Support and NOC Support together handle 41.8% of incidents but account for 79.6% of the ₹310.36 Crore total cost-to-serve. Vendor Support alone drives 21.3% of cost from just 6.3% of incidents, at an average of ₹4,17,953 per incident.

Discovery 2 — Optical Network is both the largest operational cost driver and a top SLA-breach contributor.

Optical Network incidents generate ₹2.49 Crore in operational (dispatch) cost — more than any other fault category — and together with Access Equipment account for 82.7% of all SLA breaches (2,065 of 2,497) from 45.1% of incident volume. Within Optical Network alone, 19.74% of incidents breach SLA.

Discovery 3 — Field-dispatch resolution paths, not fault severity, explain most SLA breaches.

Fiber Team Dispatch and Field Visit — the two resolution strategies requiring a physical site visit — account for 96.9% of all SLA breaches (2,420 of 2,497), despite representing only 47.0% of total incidents. Every other resolution strategy combined contributes the remaining 3.1%.

Discovery 4 — Resolution strategy cost varies by more than 800x for the same incident.

Resolution Strategy	Avg. Cost / Incident
Remote Resolution	₹515
Remote Configuration	₹11,728
Field Visit	₹48,949
NOC Support	₹1,69,672
Fiber Team Dispatch	₹2,25,026
Vendor Support	₹4,17,953
Hardware Replacement	₹4,21,540

Average cost per incident ranges from ₹515 for Remote Resolution to ₹4,21,540 for Hardware Replacement. Vendor Support resolving Core & IP faults specifically averages ₹10,00,124 (roughly ₹10 Lakh) per incident, the highest of any fault-and-resolution combination in the dataset.

Discovery 5 — Escalation, not fault severity or category, is the strongest observed predictor of SLA breach.

Escalated incidents account for 17.0% of total volume but breach SLA at 53.27%, compared with just 1.11% for non-escalated incidents—a 48-fold increase. Incidents that both breached SLA and were escalated also recorded the highest average cost, at ₹2,11,247 per incident.

Discovery 6 — Customer impact concentrates even more sharply than cost.

Core & IP and Optical Network faults together account for 63.8% of the 1.99 Crore cumulative customer-impact instances recorded, though a Core & IP incident affects 3,301 customers on average — 3.5 times an Optical Network incident's 953. Tier 1 sites, 55.8% of incidents, account for 76.6% of total customer impact, at nearly 6 times the average impact per incident of Tier 3 sites.

10. Recommendations

R1 (→Discovery 1): Direct cost review toward Fiber Team Dispatch, Vendor Support and NOC Support specifically, not incident volume overall. These three strategies cover 41.8% of incidents and 79.6% of cost — the highest-leverage place to look for cost efficiency.

R2 (→Discovery 2): Prioritize preventive maintenance investment for Optical Network infrastructure ahead of the other 8 fault categories. It is simultaneously the largest operational cost driver and, with Access Equipment, the largest SLA-breach contributor.

R3 (→Discovery 3): Review dispatch scheduling and SLA targets specifically for Fiber Team Dispatch and Field Visit, since these two strategies alone generate 96.9% of breaches. A blanket SLA policy across all resolution types does not reflect where breach risk actually sits.

R4 (→Discovery 4): Audit why Vendor Support is selected for the incidents it currently handles. At ₹4,17,953 per incident against Fiber Team Dispatch's ₹2,25,026, confirming which Vendor Support cases could shift to a lower-cost path is a direct route to cost reduction.

R5 (→Discovery 5): Treat escalation as an early-warning trigger for proactive intervention, not only a status label. Given its 48-fold association with SLA breach, earlier action on escalation-flagged incidents is likely to move both SLA and cost outcomes together.

R6 (→Discovery 6): Weight site criticality into incident response prioritization directly, alongside severity. Tier 1 sites already drive 76.6% of customer impact from 55.8% of incidents; treating them as a distinct response tier targets the incidents with the largest customer consequence.

11. Expected Business Impact

The analysis shows that operational outcomes are influenced more by resolution routing and escalation decisions than by fault type alone.

Resolution strategy choice is tied to a wide cost range that fault category alone does not explain. Vendor Support averages ₹4,17,953 per incident against Fiber Team Dispatch's ₹2,25,026 and Remote Resolution's

₹515 — the same network, three different average costs depending on which path an incident is routed through. Vendor Support alone used in 6.3% of incidents accounts for ₹66.04 Crore of the ₹310.36 Crore total cost-to-serve.

Escalation is a stronger observed predictor of SLA outcome than fault severity or fault category: escalated incidents breach SLA at 53.27% against 1.11% for incidents that are not escalated. Incidents that breach SLA without prior escalation cost an average of ₹2,11,247 — close to double the ₹1,07,236 average for incidents with neither escalation nor breach.

> ***The analysis indicates that improving incident routing and escalation decisions offers the greatest opportunity to reduce both SLA breaches and operational cost, beyond improvements targeted at fault categories alone.***